

COLLEGE STATION EASTWOOD, TX, USA

WMO: 747460

Lat: 30.589N

Lon: 96.365W

Elev: 305

StdP: 14.53

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.6	31.9	14.3	11.5	38.9	18.5	14.1	41.7	24.4	55.5	21.5	55.1	7.9	350	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.5	99.9	75.5	98.1	75.6	96.0	75.7	79.8	90.4	78.8	88.9	78.2	88.0	8.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	144.0	82.7	76.6	140.4	82.1	75.8	136.5	81.4	43.4	89.7	42.5	89.2	41.7	88.1	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.8	18.7	16.9	DB	22.9	103.3	3.7	3.1	20.3	105.5	18.1	107.4	16.1	109.1	13.4	111.4
			WB	20.7	80.6	3.5	1.2	18.2	81.4	16.2	82.1	14.3	82.8	11.7	83.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.5	51.7	55.5	61.8	68.8	76.5	82.6	85.0	85.5	80.4	71.1	60.9	53.2	(6)
	DBStd	14.15	9.93	10.29	8.96	7.02	5.48	3.59	2.89	3.27	5.13	7.54	9.09	9.66	(7)
	HDD50	263	99	56	16	1	0	0	0	0	0	0	17	74	(8)
	HDD65	1526	425	291	166	42	3	0	0	0	1	33	181	384	(9)
	CDD50	7376	151	211	382	566	820	977	1084	1100	912	655	344	174	(10)
	CDD65	3165	13	26	67	157	358	527	619	635	463	223	58	19	(11)
	CDH74	33672	47	128	365	1150	3315	5959	7649	7907	4831	1889	359	73	(12)
	CDH80	15960	2	21	50	278	1295	2911	4080	4319	2304	642	53	5	(13)
(14) Wind	WSAvg	7.8	8.3	8.9	9.1	9.3	8.8	7.6	6.8	6.4	6.3	7.0	7.6	7.8	(14)
(15) Precipitation	PrecAvg	40.9	3.2	2.9	2.7	3.4	4.8	3.9	2.3	2.3	4.6	4.1	3.2	2.8	(15)
	PrecMax	61.0	15.6	9.8	6.1	8.8	11.4	12.6	7.0	10.6	10.3	12.9	9.3	8.7	(16)
	PrecMin	28.6	0.2	0.1	0.3	0.1	0.6	0.1	0.0	0.1	0.5	0.4	0.2	0.2	(17)
	PrecStd	9.4	2.9	2.1	1.6	2.2	2.9	3.4	1.8	2.1	2.8	2.9	2.1	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.9	82.5	84.5	90.3	95.1	100.5	101.5	103.0	100.5	92.3	84.5	80.2	(19)
		MCWB	64.6	65.0	67.7	72.4	75.2	75.7	74.8	74.7	73.4	72.8	71.0	68.4	(20)
	2%	DB	74.6	77.8	81.1	85.8	91.9	97.1	99.3	100.2	96.5	89.4	81.4	75.6	(21)
		MCWB	63.6	64.8	67.3	70.6	74.7	75.1	76.0	75.4	74.3	73.2	69.5	66.3	(22)
	5%	DB	71.4	74.1	78.5	83.0	89.8	94.3	97.3	98.4	93.5	86.7	78.5	72.7	(23)
		MCWB	63.5	64.1	66.3	69.5	74.2	75.7	76.1	75.7	74.0	71.8	67.5	65.6	(24)
	10%	DB	68.3	71.1	75.1	80.8	87.3	92.2	95.1	96.2	91.1	83.7	75.1	69.8	(25)
		MCWB	61.4	63.5	64.7	68.3	73.4	75.8	76.1	75.9	74.1	70.1	67.1	64.3	(26)
	0.4%	WB	69.4	70.3	71.9	76.0	79.6	80.8	81.4	80.3	79.9	78.1	74.7	71.4	(27)
		MCDB	72.9	75.8	78.8	83.5	88.8	92.0	94.9	90.0	88.4	85.1	80.0	75.4	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	67.4	68.7	70.3	73.9	77.4	79.4	79.4	79.2	78.4	76.6	72.6	69.4	(29)
		MCDB	72.2	74.1	76.4	81.1	86.5	89.5	91.5	89.3	86.7	83.5	77.3	73.7	(30)
	5%	WB	65.8	67.0	69.0	72.6	76.3	78.3	78.4	78.4	77.6	75.4	70.8	67.5	(31)
		MCDB	70.1	71.8	74.6	79.4	84.9	87.7	89.5	88.6	86.0	81.9	75.8	71.6	(32)
	10%	WB	62.9	65.3	67.5	71.3	75.2	77.4	77.8	77.8	76.8	74.0	68.8	65.0	(33)
		MCDB	67.0	69.9	72.8	77.8	83.3	86.3	88.3	88.2	85.1	80.1	73.9	68.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	19.2	19.7	20.0	18.9	18.4	19.8	20.5	19.8	20.6	20.0	19.4	(35)
		MCDBR	22.7	21.8	20.5	20.6	20.5	21.3	22.9	23.7	22.7	21.5	21.5	21.2	(36)
		MCWBR	14.1	12.7	10.1	9.4	6.9	5.2	5.1	5.0	6.2	8.6	11.7	13.9	(37)
	5% WB	MCDBR	18.6	17.6	17.2	16.9	17.5	17.7	19.7	19.9	18.3	17.3	18.2	18.4	(38)
		MCWBR	14.2	12.4	10.9	9.3	7.4	6.0	5.5	5.2	6.1	8.7	12.3	14.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.326	0.345	0.368	0.416	0.465	0.450	0.467	0.457	0.425	0.374	0.367	0.327	(40)
	taud		2.503	2.461	2.398	2.275	2.181	2.276	2.257	2.281	2.326	2.450	2.408	2.510	(41)
	Ebn at Noon		287	291	291	279	265	267	262	264	269	277	269	281	(42)
	Edh at Noon		28	32	36	43	48	43	43	42	39	32	31	27	(43)
(44) All-Sky Solar Radiation	RadAvg		917	1034	1396	1701	1837	2025	2052	1936	1630	1349	1002	804	(44)
	RadStd		90	196	130	123	139	153	152	113	147	177	110	101	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	+155

Nomenclature: See separate page